

PS-SSR 6Q Evaluation Report

Qwen3-8B (bf16, A100) · 6 original survey questions · 66 clusters from ~12K social media posts · Mean JS divergence
Baseline to beat: Direct SSR = **0.0277**. (23Q expanded set is synthetic; quarantined.)

Table 1: Lead results: methods beating Direct SSR (`improve_orig6.json`, 2026-04-21)

Method	Mean JS ↓	Δ vs SSR	Notes
SSR + LLM-dist, $w=0.5$	0.0257	−7.2%	Primary contribution
SSR + LLM-dist, $w=0.6$	0.0258	−6.9%	LLM-only distribution estimation
SSR + LLM-dist, $w=0.7$	0.0259	−6.5%	
LLM-dist top-10 (alone)	0.0259	−6.5%	
SSR + LLM-dist, $w=0.8$	0.0261	−5.8%	
SSR top-5 clusters	0.0272	−1.8%	Tuning knob, small effect
Ensemble $w=0.95$ (SSR + ML steer)	0.0272	−1.8%	Steering as ensemble member
SSR $n=50$ posts	0.0275	−0.7%	Post-count ablation
<i>Direct SSR baseline</i>	<i>0.0277</i>	—	

Table 2: SSR + Multi-Layer Steering Ensemble (`persona_methods.json`)

Method	Div	JS ↓	Δ
BL: Direct SSR	0.0105	0.0276	—
E: Ens $w=0.5$	0.0083	0.0261	−5.4%
E: Ens $w=0.7$	0.0071	0.0264	−4.3%
E: Ens $w=0.9$	0.0088	0.0267	−3.3%
B: LLM-dist est.	0.0162	0.0273	−1.1%
C: SSR+LLM corr.	0.0134	0.0296	+7.2%
D: Proj SSR $s=0.5$	0.1214	0.0405	+46.7%
A: LLM voting	0.0594	0.2360	+755%

Steering alone fails; useful only in ensemble.

Table 3: Multi-Layer Steering Depth (`improve_orig6`)

Variant	Mean JS ↓
ML L[12,16,20,24] $n=5$	0.0643
ML L[16,20,24,28] $n=5$	0.0434
ML L[20,24] $n=5$	0.0404
ML L[16,24] $n=5$	0.0378
MultiLayer $n=10$	0.0384
MultiLayer $n=5$	0.0396

Deeper stacks hurt; 2-layer [16,24] is best in isolation but still > SSR.

Table 4: Why single-layer steering fails (`fix_steering`)

Variant	Div	JS ↓
Direct SSR	0.0105	0.0276
Multi-layer $\alpha=0.1$	0.0222	0.0298
ICL (5 posts)	0.0052	0.0368
Contrastive prompt	0.0086	0.0415
ICL + steer $\alpha=0.3$	0.0065	0.0420
Large $\alpha=3.0$, unit-norm	0.0046	0.0622
Single-layer $\alpha=0.1$, L24	0.0046	0.0659
Large $\alpha=1.0$, unit-norm	0.0037	0.0689

Single-layer steering is 2.4× worse than Direct SSR.

Table 5: Post & Cluster Ablation (`improve_orig6`)

Axis	Setting	Mean JS
Posts/cluster	30	0.0277
	50	0.0275
	80	0.0276
Top- K clusters	5	0.0272
	10	0.0272
	15	0.0277
	20	0.0280

Tuning SSR in isolation gives only marginal gains.

Conclusions. (1) **Primary result:** SSR + LLM-dist at $w=0.5$ reaches Mean JS = **0.0257**, a **7.2% relative reduction** over Direct SSR (0.0277). (2) **Two signal sources:** (i) the LLM-dist mixture accounts for most of the gain; (ii) the SSR + multi-layer steering ensemble contributes an independent ~5% improvement. (3) **Negative results to preserve:** single-layer steering at L24 (JS=0.0659), per-post LLM voting (JS=0.2360), persona-projected SSR, and deep multi-layer stacks all underperform the baseline. (4) **Proposed paper framing:** SSR + LLM-dist as the main method; multi-layer steering ensemble as ablation; drop QAW, KL demographic reweighting, and the 23Q expansion — these were evaluated on synthetic questions and do not carry over to the real benchmark.